

Kill Dozer

ELEC3020 Project Design Report

Team-27

Signed by:

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Introduction

The objective of the project is to create a SumoBot using the ESP32 TTGO microcontroller. The robot is designed to detect, engage, and push other robot opponents outside a circular ring, while remaining within the white boundary line. The project enforces practical implementation of the concepts learnt in the Embedded Systems (ELEC3020) unit.

This report documents the complete design, development, and implementation of the custom built SumoBot “KillDozer”.

Team Contributions

The parts that each team member worked on is below:

Table 1: Team Member Contributions

Team Member (Student Number)	Contributions
Sebastian Gazey (23417131)	Design Decisions Project Design Report Software (both strategy and writing the code) Chassis/Parts Design Electronics (purchasing, research, design, assembly/soldering) Assembly 3D printing Testing Bill of Materials
Mia O’Dea (23729581)	Design Decisions Chassis/Parts Design Project Design Report Electronics (soldering, design) Assembly 3D printing Testing Bill of Materials
Carla Schultz (23854485)	Bill of Materials User Manual Marketing Document Project Design Report
Sophie Mattingly (23771898)	Bill of Materials User Manual Marketing Document Project Design Report

Design Choices

There were a few key principles that KillDozer was built around, these being maintaining traction, detection (of opponents and the competition area), and flipping/pushing the opponent.

Traction

A large surface area was crucial to maintaining traction, and as such, KillDozer uses two tracks as the driving system. These had the added benefit of increasing the agility of KillDozer, as differential steering could be used, allowing it to turn on the spot - this system also had the added benefit of reducing the complexity of the steering system (as a separate servo to control a steering axle was not needed).

To increase traction even more, KillDozer was going to have Dycemtm attached to the tracks (however, the team ran out of time to implement this).

Detection

KillDozer uses two URM09 ultrasonic sensors as the main way to detect opponents. These were chosen due to having a wider field of view than HC-SR04's.

There are also RGB colour sensors at the front two corners, to detect if the robot is about to drive out of the circle.

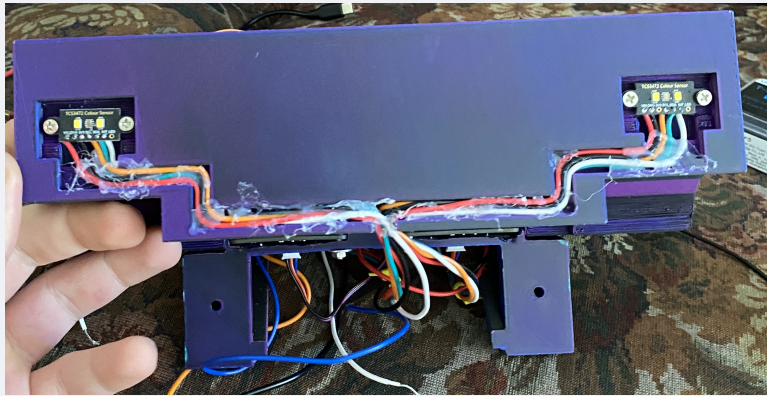


Figure 1: RGB sensors underneath the blade

Pushing/Flipping

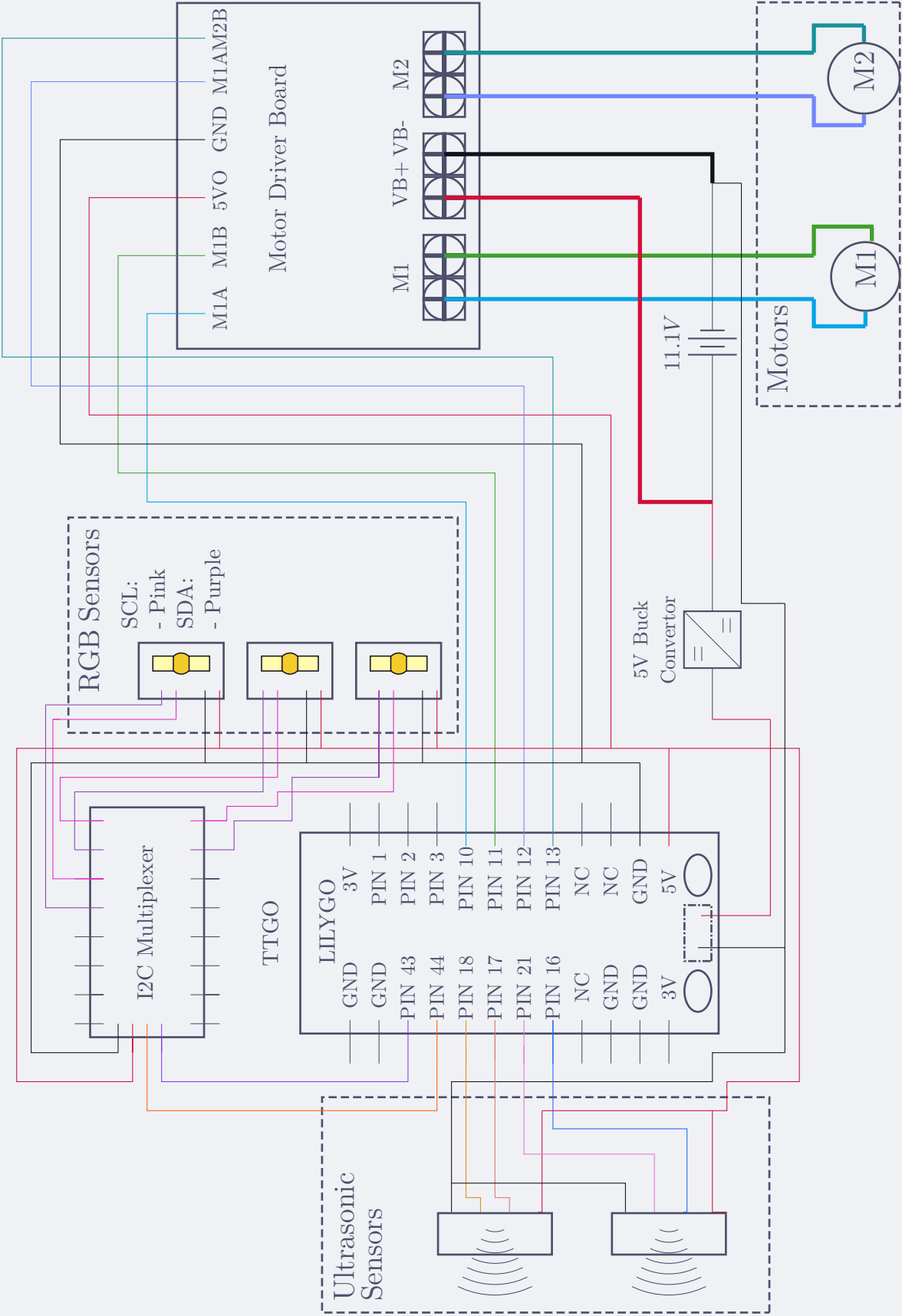
For this KillDozer has a bulldozer like blade - this has gone through numerous iterations to save weight, the final version:



Figure 2: Bulldozer blade, with URM09 sensors

Hardware:

Circuit Diagram:



Chassis:

3D-printed with PLA, as we thought it would help us stay underweight, and allowed us to designed complex parts that would be difficult to machine. PLA also is cheap, which was useful for us.

The chassis went through multiple iterations, mostly due to our initial robot melting in Week 12 (turns out, don't leave your AliExpress LiPo's in your robot while you charge them). Hence, our marketing does look a little different to our final robot. In the last week, we repurposed the robot we built for testing software on, and modified the chassis from that to fit the blade.

This weighed too much, and trying to remove excess material to lighten the robot resulted in it literally tearing itself apart at 23:00 on Wednesday night in Week 12.

Sensors:

Table 2: Reasoning behind sensor choices for KillDozer

Component	Reason
URM-09 Ultrasonic sensors	Chosen as it provides a wider cone of "view" than a regular HC-SR04, meaning KillDozer needed less ultrasonic sensors. Ultrasonic sensors were chosen over ToF sensors or infrared sensors due to being less impacted by external light sources.
RGB Sensors	Found to provide better responses than regular line sensors, and were more reliable Did require an I2C multiplexer due to the addresses being non-changeable
ToF Sensor	While we found these to be less reliable than ultrasonic sensors, once we set it inside of the blade (so it had a cover from overhead lighting), we found it was accurate enough to determine if there was a robot inside of the bulldozer blade, however this did not end up being successfully implemented in the code/strategy due to running out of time.

Software:

Software Design Diagram

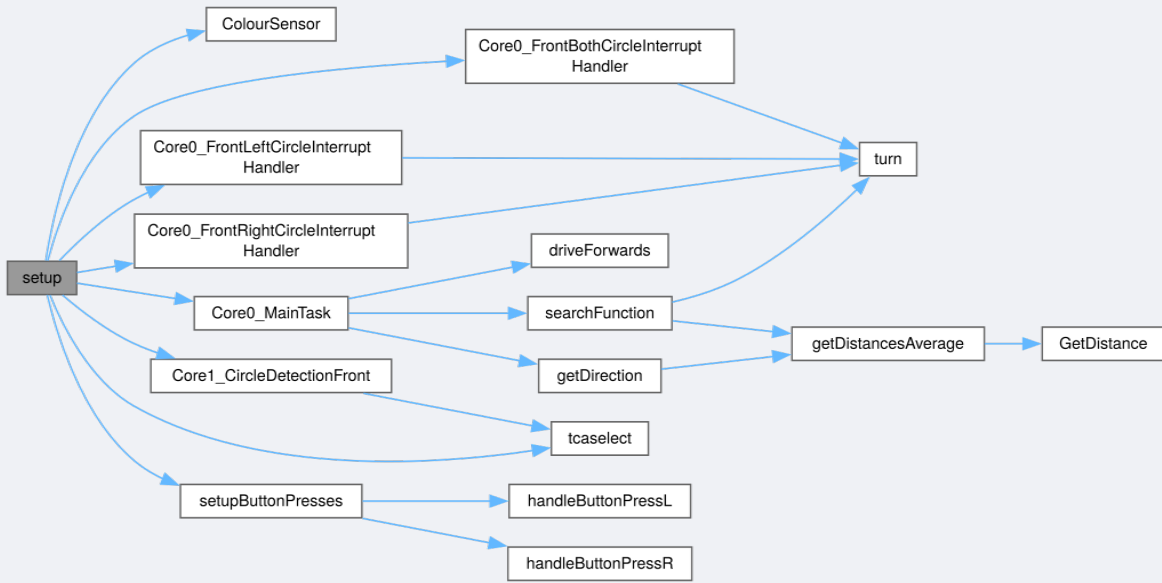


Figure 3: Software Call Graph

Strategy

KillDozer uses both of the TTGO's cores, with one (core 1) checking the RGB sensors if the robot is nearing the edge of the circle, and the other (core 0) doing everything else. Core 1 will interrupt the first core's task(s), prioritising moving the robot to safety/telling the robot where to move to next.

The main core has a simple program loop - when it starts, it first calls the `searchFunction()`, which makes the robot turn until it detects an object closer than 1.6 meters away. Once this happens, the function returns, and then the main function loops over a set of instructions that tells the robot to drive forwards (if one sensor is closer to the opponent than the other, then it drives forwards with a slight bias towards that direction). This continues until the RGB sensors detect the edge, at which point Core 1 interrupts this task (and sets a flag to break the while loop).

The interrupt handler running on Core 0 then handles the movements for the robot to take - if only one sensor is detecting the edge of the circle, the robot turns towards that sensor, until both sensors are touching (this is all done by Core 1 interrupts, using the sensor, so only one core is interfacing with the I2C bus). Turning so both are detecting the edge of the circle is done so that the opponent is definitely pushed out of the circle. Once both sensors are at the edge, the robot turns around 90 degrees, and restarts its main looping task on Core 0, searching for another opponent.

We ran out of time to implement the ToF sensor, but this would have been used to detect if the opponent was inside of the bulldozer blade.

Using both cores meant that the robot could react immediately to a detection of going outside of the circle, whereas if we were using a single core, we could still use the task based approach to allow for

interrupts, but we would not always be able to be checking the sensors, meaning that the robot could possibly go outside of the circle before it could react.

```
void Core1_CircleDetectionFront(void *pvParameters) {
    for (;;) {
        float rFR, gFR, bFR, rFL, gFL, bFL;

        float totalFR, totalFL;
        tcselect(FRONT_RIGHT_TCS_I2C_NUMBER);
        tcs_FR.getRGB(&rFR, &gFR, &bFR);

        tcselect(FRONT_LEFT_TCS_I2C_NUMBER);
        tcs_FL.getRGB(&rFL, &gFL, &bFL);
        totalFR = rFR + bFR + gFR;

        totalFL = rFL + bFL + gFL;

        if (totalFR < 350 && totalFL < 350) { // both are probably not white

            BaseType_t xHigherPriorityTaskWoken = pdFALSE;
            xTaskNotifyFromISR(interruptHandlerHandleBoth, 0, eNoAction,
                              &xHigherPriorityTaskWoken);

            if (xHigherPriorityTaskWoken)
                portYIELD_FROM_ISR();
        } else if (totalFR < 350) {

            BaseType_t xHigherPriorityTaskWoken = pdFALSE;
            xTaskNotifyFromISR(interruptHandlerHandleFR, 0, eNoAction,
                              &xHigherPriorityTaskWoken);

            if (xHigherPriorityTaskWoken)
                portYIELD_FROM_ISR();
        } else if (totalFL < 350) {

            BaseType_t xHigherPriorityTaskWoken = pdFALSE;
            xTaskNotifyFromISR(interruptHandlerHandleFL, 0, eNoAction,
                              &xHigherPriorityTaskWoken);

            if (xHigherPriorityTaskWoken)
                portYIELD_FROM_ISR();
        }

        vTaskDelay(20);
    }
}
```

Listing 1: The task running on Core 1, which handles the edge detection sensors/logic

Bill Of Materials

Table 3: Reasoning behind sensor choices for KillDozer

Model Number	Part Name	Supplier	Quantity	Price per part (\$)	Cost (\$AUD)
MDD3A	DUAL CHANNEL Motor Driver Board	Cytron	1	8.5	8.5
ZGY370	Worm Gear Motor	Aliexpress	2	4.49	8.98
TCS34725	RGB Colour Sensor	Aliexpress	2	2.4	4.8
TCA9548A	I2C Multiplexer Board	Aliexpress	1	2.14	2.14
T100/T400	Caterpillar Chain Track Pedrail Wheel	Aliexpress	2	1.55	3.1
URM09	Ultrasonic distance sensor	DFROBOT	2	7.9	15.8
VLX530	Time of Flight (ToF) Laser Ranging Sensor	Aliexpress	1	1.55	1.55
PCB	Prototype PCB Board 9x15cm Single-Sided	Aliexpress	1	0.31	0.31
OPY FLEX1K01	Filament	Aliexpress	1	3.9	3.9
M3	Stainless Steel Cap Head bolts (10pcs)	Maker Store	1	0.252	0.252
IBUW	Dupont Jumper Wire Kit 40-pin 20cm	Aliexpress	1	0.517	0.517
FINAL PRICE					49.8

INSTRUCTION MANUAL

Kill Dozer



WARNING:

CHOKING HAZARD – Small Parts.

Key Features

- Motor – driven
- Drive within black circular area limited by white line
- Push other robots out of area
- Ultrasonic and ToF distance sensor to detect other robots
- TCS34725 RGB colour sensor to not exit white line area
- Hardware modifiable by user

IMPORTANT: Read Instructions carefully

Note: The manual is intended for users with a basic understanding of electrical and mechanical engineering. Failure to comply with the manual may cause damage to the device or user.

Package Contents

- 1 x User manual
- 1 x Sumo Bot
- 1 x Lithium Polymer (LiPo) battery
- 1 x ESP32 TTGO microcontroller

Specifications

- Robot dimensions (L x W x H) (cm): 25 x 20 x 25
- Weight (kg): 1
- Maximum Speed (m/s): 10
- Battery capacity: 11.1V 2600mAh

SAFETY PRECUATIONS

- **DO NOT** pick up the robot while it is moving.
- Keep hands, hair, and loose clothing away from the moving parts when the power is on.
- Always use fully charged batteries.
- **DO NOT** operate the robot on streets. Avoid contact with people, animals, or household objects.
- **DO NOT** enter the operating area of the robot while it is switched on.
- **DO NOT** submerge the robot into water as this may damage its electrical components.
- Make sure all sensors are clean and unobstructed.
- Avoid placing the robot on uneven or unstable surfaces.

CONTROL SPECIFICATIONS

MOTOR SPEED

The Speed is dynamically adjusted based on the distance measured by the ultrasonic sensor. Distances between 10 and 100cm are linearly mapped to an integer speed value between 30 and 255 using pulse width modulation (PWM).

If the objects are close by, the robot slows down.

If objects are further away the robot accelerates.

In competitive mode, when the opposing robot is at a speed of 50 or more, the robot overrides the mapped speed and sets its speed to maximum (255) to initiate charge.

STEERING

The ultrasonic sensor is used to trigger steering behavior. If an obstacle is 15cm or more from the robot it drives forward at a mapped speed.

TROUBLESHOOTING		
ISSUE	POSSIBLE CAUSES	SOLUTION
Robot does not switch on or powers off during operation.	-Battery is uncharged, damaged, or not properly connected. -Faulty internal power wiring.	-Fully charge or replace the battery. -Check internal power connections.
Robot does not respond to commands.	-Loose or damaged hardware connections. -Logic error or bug in software.	-Secure all hardware connections. -Review and debug software control logic.
Robot moves erratically or inconsistently.	-Battery is damaged or low. -Motor malfunction.	-Fully charge or replace the battery. -Inspect motors.
Robot fails to detect obstacles.	-Ultrasonic sensor is misaligned, obstructed. -Sensor interference from ambient noise, temperature, or lighting.	-Clean sensor surface. -Recalibrate or replace the sensor.

MAINTENANCE AND CARE

MAINTENANCE AND STORAGE REQUIREMENTS

- Clean the robot regularly using a dry microfiber towel.

Ensure all sensors are clean and unobstructed.

- Store robot indoors in a dry environment with temperatures between 10°C to 30°C.
- Avoid exposure to heat, moisture, or humidity.

BATTERY SPECIFICATIONS

CHARGING

Only use a Lithium Polymer (LiPo) – specific charger. If possible, remove the battery from the robot before charging. Never leave the battery unattended while charging to avoid overcharging as it might damage the battery or start a fire. For safe charging, use a 1C charging rate, which fully charges the battery in one hour. To maintain battery performance, use the

battery shortly after charging. If the battery will not be used for a while, discharge it to the recommended storage voltage which is 3.7V per cell for a LiPo 11.1V battery.

REPLACEMENT

A typically LiPo battery has a lifespan of approximately 300 charges, through this can vary depending on usage, charging habits and care. Immediately replace the battery if it becomes hot, swollen or visibly damaged. These are indicators that the battery is no longer safe to use. Batteries in this condition should be correctly disposed of according to e – waste and hazardous waste guidelines. Additionally, many LiPo chargers include an internal resistance check feature. If the internal resistance varies significantly between cells, this may indicate cell imbalance, and the battery should be replaced.

1. Power off the robot
2. Carefully disconnect the existing battery.
3. Dispose of the old battery following LiPo battery regulations.
4. Insert a new, fully charged battery ensuring correct polarity (match the + to + terminals and – to -).

WARRANTY

A warranty claim can be made on manufacturing faults within 12 months from the date of purchase. If the claim is approved, necessary replacement parts will be provided with no cost to the user.

The warranty does not cover the following:

- Damage resulting from wear and tear.
- Damage caused by Improper use, mishandling or failure to follow the user manual.
- Damage incurred during unauthorized maintenance or repair.
- Issues related to software.

How to submit a warranty claim:

1. Provide valid proof of purchase.
2. Describe the defect in detail, including photos or videos if possible.
3. Contact customer support.

OPERATING INSTRUCTIONS

1. Connect robot to a full charged LiPo battery or USB power supply.
2. Place the robot in a clear operating area, free from unintended obstacles.
3. Press the right button on the TTGO controller to start the system.
4. Wait for the start - up message to appear on the TTGO LCD screen before operating.



KillDozer

ESP32 - Controlled SumoBot Robot

Description: The KillDozer SumoBot Robot is designed to challenge other robots in a 1m circular arena. The robot can detect opponents and push them out of the arena while remaining within a white boundary line.

Features: ESP32 TTGO microcontroller, two motors, Ultrasonic sensor, RGB colour sensor

Specifications:

Size: (L*W*H): 25cm x 20cm x 25cm

Battery: 11.1V 2600mAh

Weight: 1 kg